

Tutorial 6: CS 215

October 26, 2025

1. In class, we have studied the chi-square distribution. In slides 113 and 114 of “Families of Random Variables”, we made an argument that the sample variance obeys a chi-square distribution with $n - 1$ degrees of freedom where n is the number of samples $\{X_i\}_{i=1}^n$. The derivation for the chi-square distribution in those slides made some assumptions, and proving them was postponed as that proof required multivariate statistics. We will complete that gap here. Let $X_1, \dots, X_n$ be iid random variables each having a normal distribution with mean μ and variance σ^2 . Then we define $\bar{X} := \sum_{i=1}^n X_i/n$ and sample variance $S^2 := \sum_{i=1}^n (X_i - \bar{X})^2/(n - 1)$. Then, we want to prove that $\bar{x}$ and S^2 are independent. We will also prove that $(n - 1)S^2/\sigma^2$ is a chi-square random variable with $n - 1$ degrees of freedom.

Solution: The solution can be found in Theorem 3.6.1 from the book ‘Introduction to Mathematical Statistics’ by Hogg, Mackean, Craig. This book can be found online at <https://minerva.it.manchester.ac.uk/~saralees/statbook2.pdf>. The equation (3.6.14) requires some more explanation. The two terms on the RHS of (3.6.13) are independent. Hence the MGF of their sum is equal to the product of their individual MGFs. The MGF of the LHS of (3.6.13) is $(1 - 2t)^{-n/2}$ as V is a chi-square random variable with n degrees of freedom. Hence the MGF of the first term on the RHS of (3.6.13), i.e. the MGF of $(n - 1)S^2/\sigma^2$, equals $(1 - 2t)^{-n/2} \times (1 - 2t)^{-1/2} = (1 - 2t)^{-(n-1)/2}$ which is the MGF of a chi-square random variable with $n - 1$ degrees of freedom.

2. We did the PCA derivation in class for one direction. Repeat the derivation for two (mutually perpendicular) directions.

Solution: Let $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ be data and define the sample mean

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i, \quad \mathbf{y}_i = \mathbf{x}_i - \bar{\mathbf{x}} \quad (i = 1, \dots, n).$$

We seek an orthonormal $d \times 2$ matrix $\mathbf{U} = [\mathbf{u}_1 \ \mathbf{u}_2]$ with $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_2$ whose column space best reconstructs the data.

The reconstructed vector is $\hat{\mathbf{y}}_i = \mathbf{U}\mathbf{U}^\top \mathbf{y}_i$. The total squared reconstruction error is

$$J(\mathbf{U}) = \sum_{i=1}^n \|\mathbf{y}_i - \hat{\mathbf{y}}_i\|^2 = \sum_{i=1}^n \|\mathbf{y}_i\|^2 - \sum_{i=1}^n \|\mathbf{U}^\top \mathbf{y}_i\|^2.$$

Thus minimizing $J(\mathbf{U})$ is equivalent to maximizing

$$\sum_{i=1}^n \|\mathbf{U}^\top \mathbf{y}_i\|^2 = \text{trace}(\mathbf{U}^\top \mathbf{S} \mathbf{U}), \quad \text{where } \mathbf{S} = \sum_{i=1}^n \mathbf{y}_i \mathbf{y}_i^\top.$$

Write the eigen-decomposition $\mathbf{S} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top$ with $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_d)$ and $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$. Let $\mathbf{A} = \mathbf{V}^\top \mathbf{U}$ (so $\mathbf{A}$ is $d \times 2$ and has orthonormal columns). Then

$$\text{trace}(\mathbf{U}^\top \mathbf{S} \mathbf{U}) = \text{trace}(\mathbf{A}^\top \mathbf{\Lambda} \mathbf{A}) = \sum_{j=1}^d \lambda_j (a_{j1}^2 + a_{j2}^2).$$

Define $r_j = a_{j1}^2 + a_{j2}^2$. Because $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_2$ we have $0 \leq r_j \leq 1$ and $\sum_{j=1}^d r_j = 2$. Hence the objective is a weighted sum $\sum_j \lambda_j r_j$ with nonnegative weights summing to 2, which is maximized by assigning $r_1 = r_2 = 1$ and $r_j = 0$ for $j \geq 3$. This choice is achieved by

$$\mathbf{A} = \begin{bmatrix} \mathbf{I}_2 \\ \mathbf{0}_{(d-2) \times 2} \end{bmatrix}, \quad \text{so} \quad \mathbf{U} = \mathbf{V}\mathbf{A} = [\mathbf{v}_1 \ \mathbf{v}_2].$$

Thus the optimal two-dimensional PCA subspace is the span of the top two eigenvectors $\mathbf{v}_1, \mathbf{v}_2$, and the maximum captured scatter is $\lambda_1 + \lambda_2$. The minimal reconstruction error is

$$J_{\min} = \sum_{i=1}^n \|\mathbf{y}_i\|^2 - (\lambda_1 + \lambda_2).$$

3. Derive the maximum likelihood estimates for the mean and covariance of a multi-variate Gaussian distribution given n iid samples $\{\mathbf{x}_i\}_{i=1}^n$.

Solution: Please see slides 61, 62 of Multivariate Gaussians for the full derivation.

4. We know that the distribution of frontal and centered human face images of some size say $q \times q$ is multivariate Gaussian. Suppose you are given $N = 10^6$ such frontal and centered human face images and are asked to design an algorithm to determine whether a given image is a frontal centered human face image or some other image. How will you do this?

Solution: We assume all the images are reshaped to form vectors $\{\mathbf{x}_i\}_{i=1}^n$ where each $\mathbf{x}_i \in \mathbb{R}^{q^2 \times 1}$. Determine the mean face image $\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i$ and the covariance matrix $\hat{\mathbf{C}} = \frac{1}{N-1} \sum_{i=1}^N (\mathbf{x}_i - \hat{\boldsymbol{\mu}})(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^\top$. These estimates $\hat{\boldsymbol{\mu}}$ and $\hat{\mathbf{C}}$ are maximum likelihood estimates of the parameters of the multivariate Gaussian. Given a new image $\mathbf{q}$, evaluate the Mahalanobis distance between $\mathbf{q}$ and $\hat{\boldsymbol{\mu}}$. This is given by $DM(\mathbf{q}, \hat{\boldsymbol{\mu}}) := \sqrt{(\mathbf{q} - \hat{\boldsymbol{\mu}})^\top \hat{\mathbf{C}}^{-1} (\mathbf{q} - \hat{\boldsymbol{\mu}})}$. The squared Mahalanobis distance is chi-square distributed with $q^2 - 1$ degrees of freedom. So we declare $\mathbf{q}$ to be a face image if $DM^2(\mathbf{q}, \hat{\boldsymbol{\mu}}) \leq \chi_{\alpha, q^2-1}^2$ and declare it to be a non-face otherwise. Here χ_{α, q^2-1}^2 corresponds to significance level α .

5. Given point pairs $\{(x_i, y_i)\}_{i=1}^n$ where $y_i = mx_i + c + \eta_i$ with $\eta_i \sim \mathcal{N}(0, \sigma^2)$. Answer the following questions:

- Derive the MLEs for m and c .
- Are they biased or unbiased?
- What is their variance?
- What is the covariance of the MLE estimates?
- What can you say about the distribution of the MLE estimates?

Solution:

- We are given data points

$$(x_i, y_i), \quad i = 1, 2, \dots, n$$

and the model

$$y_i = mx_i + c + \eta_i,$$

where the noise terms η_i are i.i.d. Gaussian:

$$\eta_i \sim \mathcal{N}(0, \sigma^2).$$

Since the η_i are independent,

$$p(y_i | x_i, m, c, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_i - (mx_i + c))^2}{2\sigma^2} \right].$$

Hence the joint likelihood is

$$L(m, c, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - mx_i - c)^2 \right].$$

The log-likelihood is

$$\ell(m, c, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - mx_i - c)^2.$$

To find the MLEs of m and c , we can equivalently minimize

$$S(m, c) = \sum_{i=1}^n (y_i - mx_i - c)^2.$$

Differentiate $S(m, c)$ w.r.t. m and c and set to zero:

$$\frac{\partial S}{\partial m} = -2 \sum x_i (y_i - mx_i - c) = 0, \quad \frac{\partial S}{\partial c} = -2 \sum (y_i - mx_i - c) = 0.$$

Simplifying gives

$$\sum x_i y_i = m \sum x_i^2 + c \sum x_i, \quad \sum y_i = m \sum x_i + nc.$$

Let

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i.$$

From the second equation,

$$c = \bar{y} - m\bar{x}.$$

Substitute into the first to obtain

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad c = \bar{y} - m\bar{x}.$$

Thus the MLEs are

$$\hat{m} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{c} = \bar{y} - \hat{m}\bar{x}.$$

(b) Since $y_i = mx_i + c + \eta_i$ and $\mathbb{E}[\eta_i] = 0$,

$$\mathbb{E}[y_i] = mx_i + c.$$

Substitute y_i into $\hat{m}$:

$$\hat{m} = m + \frac{\sum (x_i - \bar{x})\eta_i}{\sum (x_i - \bar{x})^2}.$$

Because $\mathbb{E}[\eta_i] = 0$,

$$\mathbb{E}[\hat{m}] = m.$$

Similarly,

$$\hat{c} = \bar{y} - \hat{m}\bar{x} = c + \bar{\eta} - \bar{x}(\hat{m} - m),$$

and $\mathbb{E}[\bar{\eta}] = 0$ gives

$$\mathbb{E}[\hat{c}] = c.$$

Hence both $\hat{m}$ and $\hat{c}$ are unbiased.

Define

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2.$$

(c) From the expression above,

$$\hat{m} - m = \frac{\sum (x_i - \bar{x})\eta_i}{S_{xx}}.$$

Since the η_i are independent with variance σ^2 ,

$$\text{Var}(\hat{m}) = \frac{\sigma^2}{S_{xx}}.$$

For the intercept,

$$\hat{c} - c = \bar{\eta} - \bar{x}(\hat{m} - m),$$

so

$$\text{Var}(\hat{c}) = \text{Var}(\bar{\eta}) + \bar{x}^2 \text{Var}(\hat{m}) - 2\bar{x} \text{Cov}(\bar{\eta}, \hat{m}).$$

Now,

$$\text{Var}(\bar{\eta}) = \frac{\sigma^2}{n}, \quad \text{Cov}(\bar{\eta}, \hat{m}) = 0,$$

which yields

$$\text{Var}(\hat{c}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right).$$

Also,

$$\text{Cov}(\hat{m}, \hat{c}) = -\frac{\bar{x}\sigma^2}{S_{xx}}.$$

Why is $\text{Cov}(\bar{\eta}, \hat{m}) = 0$? Recall that

$$\bar{\eta} = \frac{1}{n} \sum_{i=1}^n \eta_i, \quad \hat{m} - m = \frac{\sum_{i=1}^n (x_i - \bar{x})\eta_i}{S_{xx}}, \quad S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2.$$

Then

$$\text{Cov}(\bar{\eta}, \hat{m} - m) = \text{Cov} \left(\frac{1}{n} \sum_i \eta_i, \frac{1}{S_{xx}} \sum_j (x_j - \bar{x})\eta_j \right).$$

Due to the bilinearity/linearity of covariance and due to the independence of η_i ,

$$\text{Cov}(\bar{\eta}, \hat{m} - m) = \frac{1}{nS_{xx}} \sum_i (x_i - \bar{x}) \text{Cov}(\eta_i, \eta_i) = \frac{\sigma^2}{nS_{xx}} \sum_i (x_i - \bar{x}).$$

But

$$\sum_i (x_i - \bar{x}) = 0,$$

so we have

$$\text{Cov}(\bar{\eta}, \hat{m}) = 0.$$

Remark: The slope estimate $\hat{m}$ depends only on the deviations of the noise terms η_i weighted by $(x_i - \bar{x})$, whose weights sum to zero. The mean noise $\bar{\eta}$ represents the average offset of all η_i , which is orthogonal (uncorrelated) to these deviations. Hence the covariance is zero.

(d) We will now do the derivation of $\text{Cov}(\hat{m}, \hat{c})$. Recall that

$$\hat{c} - c = \bar{\eta} - \bar{x}(\hat{m} - m).$$

Taking covariance with $\hat{m} - m$ gives

$$\text{Cov}(\hat{m} - m, \hat{c} - c) = \text{Cov}(\hat{m} - m, \bar{\eta}) - \bar{x} \text{Cov}(\hat{m} - m, \hat{m} - m).$$

The first term is zero by the result above, and the second term is simply the variance of $\hat{m}$. Thus,

$$\text{Cov}(\hat{m}, \hat{c}) = -\bar{x} \text{Var}(\hat{m}) = -\bar{x} \frac{\sigma^2}{S_{xx}}.$$

Therefore,

$$\text{Cov}(\hat{m}, \hat{c}) = -\frac{\bar{x}\sigma^2}{S_{xx}}.$$

Remark: When the mean of the x_i values ($\bar{x}$) is positive, a larger estimated slope $\hat{m}$ typically results in a smaller estimated intercept $\hat{c}$ so that the fitted line still passes near the centroid $(\bar{x}, \bar{y})$. Hence the covariance is negative.

- (e) From earlier parts, we see that $\hat{m} = m + \frac{\sum_{i=1}^n (x_i - \bar{x})\eta_i}{S_{xx}}$ is Gaussian distributed as it is a linear combination of the independent η_i values. Hence $\hat{m} \sim \mathcal{N}(m, \frac{\sigma^2}{S_{xx}})$.

We now work on the distribution of $\hat{c}$. Let $\bar{\eta} = \frac{1}{n} \sum_{i=1}^n \eta_i$. Since $\bar{y} = m\bar{x} + c + \bar{\eta}$ and $\hat{c} = \bar{y} - \hat{m}\bar{x}$,

$$\hat{c} = c + \bar{\eta} - \bar{x}(\hat{m} - m) = c + \bar{\eta} - \bar{x} \sum_{i=1}^n a_i \eta_i = c + \sum_{i=1}^n b_i \eta_i,$$

where

$$b_i = \frac{1}{n} - \bar{x} \frac{x_i - \bar{x}}{S_{xx}}.$$

Again, $\hat{c}$ is a linear combination of independent Gaussian variables, hence it is also Gaussian with

$$\mathbb{E}[\hat{c}] = c,$$

and

$$\text{Var}(\hat{c}) = \sigma^2 \sum_{i=1}^n b_i^2.$$

Carrying out the algebra gives the standard closed form:

$$\text{Var}(\hat{c}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right).$$

Therefore,

$$\hat{c} \sim \mathcal{N}\left(c, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)\right).$$